As a first step toward resolving this puzzle, we need to understand in more concrete terms what it means for the known laws of physics to be time-reversal symmetric. To this end, imagine it's the twenty-fifth century and you're playing tennis in the new interplanetary league with your partner, Coolstroke Williams. Somewhat unused to the reduced gravity on Venus, Coolstroke hits a gargantuan backhand that launches the ball into the deep, isolated darkness of space. A passing space shuttle films the ball as it goes by and sends the footage to CNN (Celestial News Network) for broadcast. Here's the question: If the technicians at CNN were to make a mistake and run the film of the tennis ball in reverse, would there be any way to tell? Well, if you knew the heading and orientation of the camera during the filming you might be able to recognize their error. But could you figure it out solely by looking at the footage itself, with no additional information? The answer is no. If in the correct (forward) time direction the footage showed the ball floating by from left to right, then in reverse it would show the ball floating by from right to left. And certainly, the laws of classical physics allow tennis balls to move either left or right. So the motion you see when the film is run in either the forward time direction or the reverse time direction is perfectly consistent with the laws of physics.

We've so far imagined that no forces were acting on the tennis ball, so that it moved with constant velocity. Let's now consider the more general situation by including forces. According to Newton, the effect of a force is to change the velocity of an object: forces impart accelerations. Imagine, then, that after floating awhile through space, the ball is captured by Jupiter's gravitational pull, causing it to move with increasing speed in a downward, rightward-sweeping arc toward Jupiter's surface, as in Figures 6.1a and 6.1b. If you play a film of this motion in reverse, the tennis ball will appear to move in an arc that sweeps upward and toward the left, away from Jupiter, as in Figure 6.1c. Here's the new question: is the motion depicted by the film when played backward—the time-reversed motion of what was actually filmed—allowed by the classical laws of physics? Is it motion that could happen in the real world? At first, the answer seems obviously to be yes: tennis balls can move in downward arcs to the right or upward arcs to the left, or, for that matter, in innumerable other trajectories. So what's the